Let us see some more notations. We are familiar with notations like $\sum_{i=1}^{n} a_i$. This in fact stands for $a_1 + a_2 + \cdots + a_n$. Similarly we can use the notations $\bigcup_{i=1}^{n} A_i$ and $\bigcap_{i=1}^{n} A_i$ to denote $A_1 \cup A_2 \cup \cdots \cup A_n$ and $A_1 \cap A_2 \cap \cdots \cap A_n$ respectively.

Thus, $\bigcup_{i=1}^{n} A_i = \{x : x \in A_i \text{ for some } i\}$ and $\bigcap_{i=1}^{n} A_i = \{x : x \in A_i \text{ for each } i\}$. These notations are useful when we discuss more number of sets.

If A is a set, then the set of all subsets of A is called the ***power set*** of A and is usually denoted as $\mathscr{P}(A)$. That is, $\mathscr{P}(A) = \{B : B \subseteq A\}$. The number of elements in $\mathscr{P}(A)$ is 2^n, where n is the number of elements in A.

Now, to define the complement of a set, it is necessary to know about the concept of universal set. Usually all sets under consideration in a mathematical process are assumed to be subsets of some fixed set. This basic set is called the ***universal set***. For example, depending on the situation, for the set of prime numbers, the universal set can be any one of the sets containing the set of prime numbers. Thus, one of the sets $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ may be taken as a universal set for the set of prime numbers, depending on the requirement. Universal set is usually denoted by U.

To define the complement of a set, we have to fix the universal set. Let A be a subset of the universal set U. The ***complement*** of A with respect to U is denoted as A' or A^c and defined as $A' = \{x : x \in U \text{ and } x \notin A\}$.

The ***set difference*** of the set A to the set B is denoted by either $A - B$ or $A \backslash B$ and is defined as

$$A - B = \{a : a \in A \text{ and } a \notin B\}.$$

Note that,

(i) $U - A = A'$ (ii) $A - A = \emptyset$ (iii) $\emptyset - A = \emptyset$ (iv) $A - \emptyset = A$ (v) $A - U = \emptyset$.

The ***symmetric difference*** between two sets A and B is denoted by $A \Delta B$ and is defined as $A \Delta B = (A - B) \cup (B - A)$. Actually the elements of $A \Delta B$ are the elements of $A \cup B$ which are not in $A \cap B$. Thus $A \Delta B = (A \cup B) - (A \cap B)$.

A set X is said to be a ***finite set*** if it has k elements for some $k \in \mathbb{W}$. In this case, we say the finite set X is of ***cardinality*** k and is denoted by $n(X)$. A set is an ***infinite set*** if it is not finite. For an infinite set A, the cardinality is infinity. If $n(A) = 1$, then it is called a ***singleton set***. Note that $n(\emptyset) = 0$ and $n(\{\emptyset\}) = 1$.

1.2.1 Properties of Set Operations

We now list out some of the properties.

Commutative

(i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$.

Associative

(i) $(A \cup B) \cup C = A \cup (B \cup C)$ (ii) $(A \cap B) \cap C = A \cap (B \cap C)$.

Distributive

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Identity

(i) $A \cup \emptyset = A$ (ii) $A \cap U = A$.

Idempotent

(i) $A \cup A = A$ (ii) $A \cap A = A$.

Absorption

(i) $A \cup (A \cap B) = A$ (ii) $A \cap (A \cup B) = A$.